S T U D S
Learning Platform | Study Notes

COMPUTER NETWORKS

Unit 1

Introduction to Data Communication & Physical Layer

Chapter 1.1
Introduction

University-level study notes for examination preparation and semester revision

Topics covered: data communication, uses of networks, LAN / MAN / WAN, topologies, network hardware, network software, connection-oriented and connectionless services, service primitives, OSI model, TCP/IP model

Chapter Overview

This chapter lays the foundation for the whole subject. It explains what data communication and computer networks are, why networks are built, how they are classified by size, how devices are arranged (topology) and joined together (hardware), and how the software of a network is organised into layers and protocols. The chapter ends with the two reference architectures used to describe networks: the OSI reference model and the TCP/IP model.

Learning Objectives

After studying this chapter, a student should be able to:

- define data communication and a computer network, and state the uses of networks;
- distinguish LAN, MAN and WAN;
- describe bus, star, ring, mesh, tree and hybrid topologies with their advantages and disadvantages;
- explain the role of each network hardware component and distinguish similar devices (hub and switch, switch and router, router and gateway, repeater and bridge);
- explain layering, protocols and the main design issues of network software;
- compare connection-oriented and connectionless services and explain service primitives;
- list the layers of the OSI and TCP/IP models, state their functions, and compare the two models.

Syllabus Coverage Map

No.	Syllabus topic	Section in these notes
1	Introduction to Data Communication and Computer Networks	Section 1
2	Uses of Computer Networks	Section 2
3	Types of Computer Networks: LAN, MAN, WAN	Section 3
4	Network Topologies	Section 4
5	Network Hardware Components	Section 5
6	Network Software: network design issues, protocols	Section 6
7	Connection-Oriented vs Connectionless Services	Section 7
8	Service Primitives	Section 8
9	OSI Reference Model	Section 9
10	TCP/IP Model	Section 10
-	OSI vs TCP/IP (comparison)	Section 11
-	Definitions, comparisons, revision and exam questions	Final sections

Scope and source note. These notes cover only the Chapter 1.1 syllabus. Each topic is explained using the standard foundational treatment found in networking textbooks. Protocol names, device types and figures given as examples are indicative and are not exhaustive. Topics that belong to later chapters of Unit 1 (for example transmission media, switching and multiplexing) are deliberately not covered.

1. Introduction

Modern computing depends on the ability of machines to exchange information. This section defines the two central ideas of the subject, data communication and computer networks, and explains how they are related.

1.1 Data Communication

Definition

> **Data communication** is the exchange of data between two or more devices through a transmission medium such as a cable or a wireless link.

Explanation

Data communication is concerned with how data is carried from a sender to a receiver. For communication to take place, five components are required.

Component	Meaning	Example
Message	The information (data) to be communicated.	Text, numbers, images, audio, video
Sender	The device that sends the message.	Computer, mobile phone, workstation
Receiver	The device that receives the message.	Computer, printer, television
Transmission medium	The physical path over which the message travels.	Twisted-pair cable, optical fibre, radio waves
Protocol	A set of rules that governs the communication.	Without a protocol two devices may be connected but cannot understand each other

Example

When a student sends an e-mail, the student's computer is the sender, the e-mail text and attachments form the message, the Internet connection is the medium, the mail server and the recipient's computer act as receivers, and the mail protocols define the rules of the exchange.

Modes of data flow

The direction of data flow between two devices can be of three kinds.

Mode	Direction of flow	Example
Simplex	One way only. One device always transmits and the other always receives.	Keyboard to computer, television broadcast
Half-duplex	Both directions, but only one direction at a time.	Walkie-talkie
Full-duplex	Both directions at the same time.	Telephone conversation

Key Points

- Data communication is judged by four characteristics: **delivery** (to the correct destination), **accuracy** (data is not altered), **timeliness** (delivered without undue delay) and **jitter** (uneven delay in packet arrival, which should be small for audio and video).
- A protocol is essential: the five components together make up a data communication system.

1.2 Computer Network

Definition

> A **computer network** is a collection of autonomous computers interconnected by a single technology so that they can exchange information and share resources.

Explanation

The word “autonomous” means that each computer can work on its own; none of them is simply a controlled slave of another. Two computers are said to be interconnected when they are able to exchange information, whether by copper wire, optical fibre, microwave, infrared or satellite links.

A network is built from the following elements:

- **Nodes (hosts):** computers, servers, printers and other devices that send or receive data.
- **Links:** the communication paths (wired or wireless) that join the nodes.
- **Connecting devices:** hubs, switches, routers and similar devices that direct data between nodes.
- **Protocols and network software:** the rules and programs that make communication possible.

Example

The computers of a college laboratory connected to a common switch, sharing one printer and one file server, form a simple computer network.

Network criteria

The usefulness of a network is judged by three main criteria:

- **Performance:** measured by transit time, response time, throughput (amount of data delivered per unit time) and delay.
- **Reliability:** measured by how often failures occur and how quickly the network recovers from them.
- **Security:** protection of data from unauthorised access, damage and loss.

1.3 Data Communication and Computer Networks Compared

Feature	Data communication	Computer network
Focus	The process of transferring data between two devices.	The overall system of interconnected devices that supports many such transfers.
Scope	Narrow: concerned with transmission, medium and signals.	Broad: includes hardware, software, topology, protocols and services.
Question answered	How is data carried from A to B?	How are many devices connected so that they can share data and resources?
Relationship	Provides the basic transfer mechanism.	Uses data communication to provide resource sharing and services.

2. Uses of Computer Networks

Networks are used by organisations, by individuals at home, and by people on the move. The uses can be grouped into three areas, followed by the social issues that networks raise.

2.1 Business Applications

Explanation

- **Resource sharing:** programs, data and equipment such as printers and storage are made available to everyone on the network regardless of their physical location. One printer can serve a whole department.
- **Client-server model:** data is kept on powerful computers called **servers**, usually in a central machine room. Employees use simpler machines called **clients** to access it. A client sends a request and the server sends back a reply.
- **Communication medium:** e-mail, instant messaging, video conferencing and voice over IP allow employees to communicate quickly and cheaply, even between distant offices.
- **Electronic commerce:** organisations do business with customers (business-to-consumer) and with other organisations (business-to-business) over the network.
- **Reliability and saving of money:** important files can be replicated on two or more machines so work continues if one fails; small low-cost computers can replace one expensive large computer, and capacity can be added step by step.

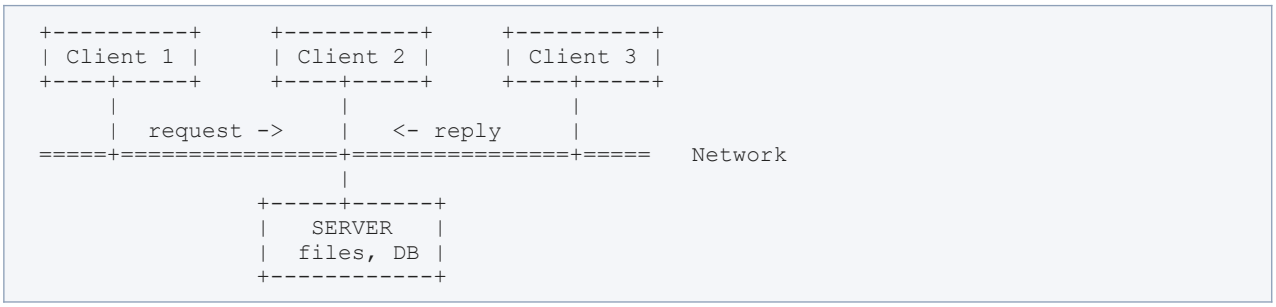


Figure 2.1: Client-server model

2.2 Home Applications

- **Access to remote information:** browsing the World Wide Web for news, education, entertainment and services.
- **Person-to-person communication:** e-mail, instant messaging, social networking and video calls.
- **Peer-to-peer communication:** users exchange files directly among themselves without a central server.
- **Interactive entertainment:** video on demand, online games and music streaming.
- **Electronic commerce at home:** online shopping, banking, bill payment and ticket booking.

2.3 Mobile Users

Laptops, tablets and smartphones allow people to use network services while travelling. Wireless networks make this possible. It is useful to separate two ideas: **wireless** means communication without cables, and **mobile** means that the user moves from place to place while using the network. A fixed office may use a wireless network without any mobility, while a moving user needs both.

2.4 Social Issues

The wide use of networks also raises concerns such as privacy of personal information, security against unauthorised access, spread of false or harmful material, and misuse of the network for fraud. These issues explain why security is treated as a network design requirement.

2.5 Summary of Uses

Area	Typical uses	Main benefit
Business	Resource sharing, client-server access, e-mail, video conferencing, e-commerce	Lower cost, central data, faster communication, reliability
Home	Web access, messaging, peer-to-peer sharing, entertainment, online banking and shopping	Information, convenience, entertainment
Mobile	Remote access to e-mail, files and services using wireless links	Access anywhere, any time

3. Types of Computer Networks

Networks are commonly classified by their **scale**, that is, by the geographical area they cover. The three classes in the syllabus are the Local Area Network (LAN), the Metropolitan Area Network (MAN) and the Wide Area Network (WAN).

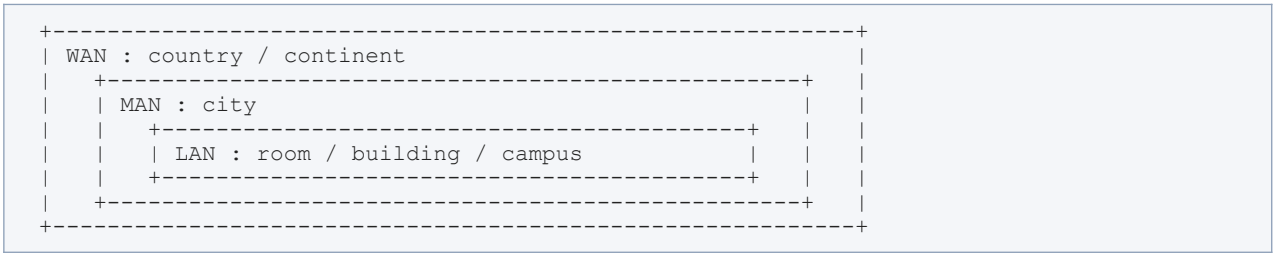


Figure 3.1: Networks classified by area covered

3.1 Local Area Network (LAN)

Definition

> A **Local Area Network (LAN)** is a privately owned network that connects computers and devices within a limited area such as a room, a building or a campus.

Explanation

A LAN is used to share resources (printers, files, Internet connection) among the users of one organisation. Because the distance is short, a LAN offers high data rates and low delay, and it is easy to manage since a single organisation owns it. Wired LANs commonly use Ethernet, and wireless LANs commonly use Wi-Fi (IEEE 802.11).

Example

A computer laboratory in a college, an office floor, or the network inside a home.

Key Points

- Small area (up to a few kilometres), privately owned.
- High speed and low error rate.
- Usually uses a single transmission technology such as Ethernet or Wi-Fi, often in bus, star or ring topology.

3.2 Metropolitan Area Network (MAN)

Definition

> A **Metropolitan Area Network (MAN)** is a network that covers a whole city or a large town, connecting several LANs within that region.

Explanation

A MAN lies between a LAN and a WAN in size. It may be owned by one organisation or may be a public utility offering service to many users. It is often used to join the offices of an organisation located in different parts of a city, or to provide high-speed access to many subscribers.

Example

A cable television network that also carries Internet service to the homes of a city is the classic example of a MAN. City-wide wireless broadband (IEEE 802.16) is another.

Key Points

- Covers roughly a city (tens of kilometres).
- Can interconnect several LANs.
- Can be private or public.

3.3 Wide Area Network (WAN)

Definition

> A **Wide Area Network (WAN)** is a network that spans a large geographical area, such as a country, a continent or the whole world, and connects many LANs and MANs.

Explanation

A WAN consists of two parts. The **hosts** are the user machines that run applications. The **communication subnet** (or simply subnet) carries messages between hosts and consists of transmission lines and switching elements (routers). Because the distances are very large, data rates are lower and delays are longer than in a LAN, and the network is usually built using lines leased from telecommunication carriers.

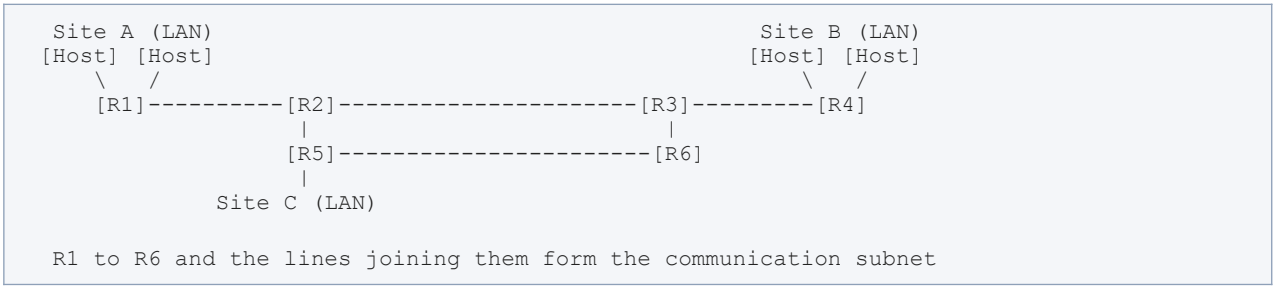


Figure 3.2: A wide area network (hosts and subnet)

Example

The network of a bank connecting its branches across the country, the network of an Internet service provider, and the Internet itself, which is a collection of interconnected networks (an internetwork).

Key Points

- Covers a very large area, often owned by several parties.
- Lower speed, higher delay and higher cost than a LAN.
- Routers in the subnet forward data towards its destination.

3.4 LAN vs MAN vs WAN

Feature	LAN	MAN	WAN
Full form	Local Area Network	Metropolitan Area Network	Wide Area Network
Area covered	Room, building, campus (up to a few km)	City or town (tens of km)	Country, continent or world
Ownership	Private, single organisation	Private or public	Several organisations or carriers; often public
Data rate	High	Moderate to high	Lower than LAN
Delay	Very low	Low to moderate	High
Error rate	Low	Moderate	Higher

Feature	LAN	MAN	WAN
Cost of setup and maintenance	Low	Moderate	High
Typical technology	Ethernet, Wi-Fi	Cable TV network, IEEE 802.16	Leased lines, routers, satellite links
Example	College laboratory	City-wide cable network	The Internet, bank branch network

4. Network Topologies

Definition

> **Network topology** is the geometric arrangement of the nodes and the links of a network, that is, the way in which devices are connected to one another.

Explanation

Two kinds of topology are distinguished. The **physical topology** describes the actual layout of the cables and devices. The **logical topology** describes the path that data follows through the network. A network can have a physical topology that differs from its logical topology.

Two types of connection are used to build a topology:

- **Point-to-point:** a dedicated link joins exactly two devices.
- **Multipoint (shared):** more than two devices share a single link.

4.1 Bus Topology

Structure

All devices are connected to a single common cable called the **bus** or **backbone**. Each device attaches to the cable through a drop line or a tap. The two ends of the cable carry **terminators** that absorb the signal and prevent it from reflecting back.

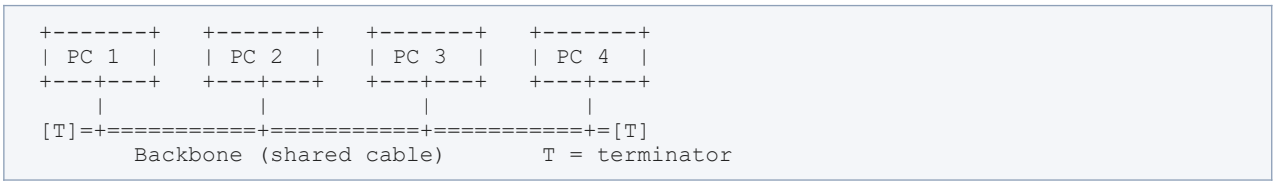


Figure 4.1: Bus topology

How communication works

A device that wants to send places its data on the bus. The signal travels in both directions along the cable and every device receives it, but only the device whose address matches the destination address accepts the data. Because the cable is shared, only one device can transmit at a time, and a method of access control is needed to handle simultaneous transmissions (collisions).

Advantages

- Easy to install and extend.
- Uses less cable than star topology.
- Low cost for a small network.

Disadvantages

- A break in the backbone cable stops the whole network or splits it in two.
- Performance falls as more devices are added, because of collisions.
- Faults are difficult to locate.

Typical use

Early Ethernet LANs (coaxial cable) and small temporary networks.

4.2 Star Topology

Structure

Every device has a dedicated point-to-point link to a central device, which is a **hub** or a **switch**. Devices are not connected directly to one another.



Figure 4.2: Star topology

How communication works

A device sends data to the central device, which forwards it to the destination. A hub repeats the data to all ports; a switch forwards it only to the port of the destination device.

Advantages

- Easy to install, expand and manage.
- Failure of one link or one device does not affect the others.
- Faults are easy to detect and isolate.

Disadvantages

- Failure of the central device stops the entire network.
- Needs more cable than bus topology, since each device has its own cable to the centre.
- Cost of the central device is an added expense.

Typical use

Most present-day wired LANs, such as office and laboratory networks built around a switch.

4.3 Ring Topology

Structure

Each device is connected to exactly two neighbours, one on each side, so that the devices form a closed loop. Each device contains a repeater function that regenerates the signal.



Figure 4.3: Ring topology

How communication works

Data travels around the ring in one direction (in a single ring). Each device receives the data, checks the destination address, and either copies it (if it is the destination) or passes it on to the next device. A control signal called a **token** is used in token ring networks to decide which device may transmit.

Advantages

- Orderly access; no collisions when a token is used.

- Performance stays fairly steady under heavy load.
- Fairly easy to install and reconfigure.

Disadvantages

- A break in the ring or the failure of one device can stop the whole network (unless a dual ring is used).
- Adding or removing a device disturbs the network.
- Data passes through every intermediate device, which adds delay.

Typical use

Token Ring LANs and FDDI backbone networks.

4.4 Mesh Topology

Structure

In a **full mesh** every device has a dedicated point-to-point link to every other device. For n devices, the number of links is n(n-1)/2 and each device needs n-1 ports.



Figure 4.4: Full mesh topology (4 devices, 6 links)

How communication works

Data goes from the sender to the receiver over the dedicated link between them. If that link fails, the data can be routed through other devices.

Advantages

- Very reliable: failure of one link does not stop communication.
- No traffic problem, since each link carries traffic only between its two devices.
- Good privacy and security because of dedicated links.

Disadvantages

- Large amount of cabling and a large number of ports, so installation and cost are high.
- Difficult to install and reconfigure as the network grows.

Typical use

Backbone networks and critical connections where reliability matters; **partial mesh** is used in WANs where only some devices have multiple links.

4.5 Tree Topology

Structure

A tree topology is a hierarchical arrangement in which groups of star networks are connected to a central root device. It combines features of the star and bus topologies.



Figure 4.5: Tree topology

How communication works

Data from a device goes up the hierarchy as far as needed and then down to the destination. A parent device controls the devices below it.

Advantages

- Easy to expand by adding new branches.
- Faults in one branch can be isolated from the rest of the network.
- Suits hierarchical organisations.

Disadvantages

- Failure of the root or of an upper-level device cuts off all the devices below it.
- Cabling and maintenance are more complex.

Typical use

Networks of a large organisation, with departments connected below a central device.

4.6 Hybrid Topology

Structure

A hybrid topology is a combination of two or more different basic topologies, for example several star networks joined together by a bus backbone.

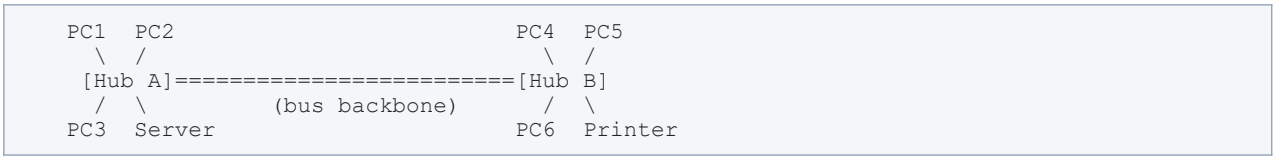


Figure 4.6: Hybrid topology (stars joined by a bus)

Advantages and disadvantages

- Flexible: each part of the network can use the topology best suited to it, and the network can grow easily.
- Design, installation and fault finding are more complex and more costly than for a single topology.

4.7 Comparison of Topologies

Feature	Bus	Star	Ring	Mesh	Tree
Connection	Common backbone	Central hub or switch	Closed loop	Every pair of devices	Hierarchy of stars
Cable required	Least	More	Moderate	Most	More
Cost	Low	Moderate	Moderate	High	Moderate to high
Effect of one link or node failure	Cable break stops network	Only that device is affected	Can stop the ring	Little effect	Branch is affected

Feature	Bus	Star	Ring	Mesh	Tree
Single point of failure	Backbone cable	Central device	Any node or link	None	Root and upper devices
Ease of expansion	Easy	Easy	Moderate	Difficult	Easy
Fault detection	Difficult	Easy	Moderate	Easy	Moderate

5. Network Hardware Components

A network is built from a number of hardware devices. They differ in what they do and in the layer of the OSI model at which they work (the layers are described in Section 9). Devices working at a lower layer are simpler and see only signals or frames, while devices working at a higher layer understand more of the data and can make better decisions.

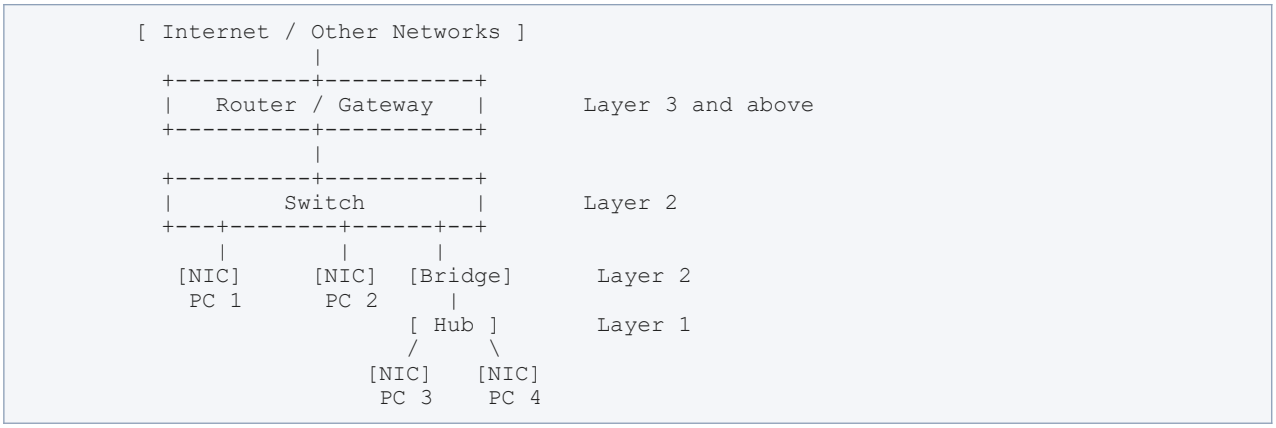


Figure 5.1: How network devices fit together (connectors and transceivers are present at every cable end)

5.1 Connectors

- **What it is:** a physical component that joins a cable to a device, or joins two cables, giving a firm electrical (or optical) contact.
- **Main purpose:** to make connection and disconnection of cables simple, reliable and standard.
- **How it works:** the connector fits into a matching socket (port) on the device or on another connector. Its shape and pin arrangement are fixed by standards so that equipment from different makers fits together.
- **Where it is used:** at the ends of every network cable, on NICs, switch ports and wall outlets.

Connector	Used with	Remarks
RJ-45	Twisted-pair Ethernet cable	Eight-pin plug; the most common LAN connector
RJ-11	Telephone line	Smaller connector with 4 or 6 positions
BNC and T-connector	Thin coaxial cable (bus LANs)	Twist-lock connector; T-connector attaches a device to the bus; a terminator closes each end
AUI (DB-15)	Older Ethernet transceiver cable	Connects a NIC to an external transceiver
SC, ST, LC	Optical fibre cable	Fibre connectors that align the fibre cores precisely

5.2 Transceivers

- **What it is:** a device that combines a **trans**mitter and a re**ceiver** in a single unit.
- **Main purpose:** to send signals onto the medium and to receive signals from it.
- **How it works:** it converts the data from the device into the form needed on the medium (electrical, optical or radio signal) and converts incoming signals back. In classic Ethernet the transceiver also senses the medium for activity and collisions.

- **Where it is used:** built into NICs and switch ports; also available as plug-in modules (for example for fibre links) and as media converters that link one type of cable to another.
- **Difference from similar devices:** a transceiver connects a device to the medium and does not extend the length of the cable; a repeater, in contrast, regenerates a signal between two cable segments.

5.3 Repeaters

- **What it is:** a device at the **physical layer** that receives a weakened signal and retransmits it at full strength.
- **Main purpose:** to extend the distance over which a signal can travel. As a signal moves along a cable it is attenuated (loses strength) and may be distorted by noise.
- **How it works:** the repeater does not amplify the noise along with the signal; it regenerates the original bit pattern and sends a fresh signal onto the next segment. It does not read addresses and does not filter data: everything it receives is passed on.
- **Where it is used:** to join two segments of the same LAN when the total cable length exceeds the limit of the medium.
- **Difference from similar devices:** an amplifier increases the strength of the signal together with the noise, whereas a repeater regenerates it. A repeater connects segments of one network; it does not separate traffic (see Bridge).

5.4 Hubs

- **What it is:** a multiport repeater working at the **physical layer**, used as the central point of a star topology.
- **Main purpose:** to connect several devices in one LAN segment.
- **How it works:** when a frame arrives at one port, the hub regenerates the signal and sends it out of **all other ports**. It does not examine addresses, so every device receives every frame and the devices share the available bandwidth. All devices attached to a hub lie in a single collision domain, and communication is half-duplex.
- **Types:** a passive hub only connects the wires; an active hub regenerates the signal; an intelligent hub adds management features.
- **Where it is used:** small, older LANs. Hubs have largely been replaced by switches.

5.5 Network Interface Cards (NICs)

- **What it is:** the hardware (a card or a chip on the motherboard) that connects a computer to the network. It is also called a network adapter or LAN card.
- **Main purpose:** to give the computer the ability to send and receive data on the network.
- **How it works:** the NIC converts data from the computer into frames and signals suitable for the medium and does the reverse for incoming signals. It carries a unique **MAC (physical) address**, a 48-bit address fixed by the manufacturer and written in hexadecimal, for example 00:1A:2B:3C:4D:5E. The NIC accepts a frame addressed to its own MAC address (or to all devices) and ignores others. It also performs frame formation, error detection and buffering.
- **Where it is used:** in every host (desktop computer, server, printer) on a wired or wireless network. A wired NIC has a port for an RJ-45 connector; a wireless NIC has an antenna.
- **Layer:** physical and data link layers.

5.6 PC Cards

- **What it is:** a credit-card-sized removable expansion card that fits into a slot of a laptop computer. The standard was defined by PCMCIA (Personal Computer Memory Card International Association). Its later 32-bit version is called CardBus.
- **Main purpose:** to add features to a laptop, which has little internal space. A network PC card gives a laptop an Ethernet, wireless LAN or modem connection.
- **How it works:** when inserted into the slot, the card acts as a NIC for the laptop. PC cards come in three types (I, II, III) that differ in thickness.
- **Where it is used:** in laptops and other portable computers. In current equipment, built-in network adapters and USB adapters have taken over this role.
- **Difference from similar devices:** a PC card is a NIC in a small removable form for portable computers, while an ordinary NIC is installed in a desktop computer.

5.7 Bridges

- **What it is:** a device at the **data link layer** that connects two LAN segments and makes them work as one network.
- **Main purpose:** to divide a large LAN into smaller segments, reduce traffic and join LANs.
- **How it works:** a bridge examines the **MAC address** in each frame. It builds a table of addresses by noting the source address of frames arriving on each port (a transparent or learning bridge). If the destination is on the same segment as the sender, the frame is **filtered** (dropped); if it is on the other segment, the frame is **forwarded**; if the address is unknown, the frame is sent to all other ports. A bridge has two ports in its simplest form.
- **Where it is used:** to join two LANs or two segments of a LAN of the same kind.
- **Difference from similar devices:** unlike a repeater, a bridge reads frames, filters traffic and creates a separate collision domain for each segment.

5.8 Switches

- **What it is:** a multiport bridge working at the **data link layer** (layer 2); the central device of present-day star LANs.
- **Main purpose:** to connect many devices and to forward each frame only to the device that needs it.
- **How it works:** the switch learns the MAC address of the device on each port and keeps them in a MAC address table. For each arriving frame, it looks up the destination address and sends the frame out of that one port only. If the address is not known, the frame is flooded to all ports except the incoming one. Each port is a separate collision domain, and ports can work in full-duplex mode, so many pairs of devices can communicate at the same time.
- **Forwarding methods:** store-and-forward (the whole frame is received and checked before forwarding) and cut-through (forwarding starts as soon as the destination address is read).
- **Where it is used:** in nearly all wired LANs. A layer-3 switch also carries out some routing functions.

5.9 Routers

- **What it is:** a device at the **network layer** (layer 3) that connects two or more different networks.
- **Main purpose:** to forward packets from one network to another and to select the best path to the destination.

- **How it works:** the router reads the **logical (IP) address** in each packet, looks it up in its **routing table**, and sends the packet to the next router or to the destination network. The routing table is built and updated with the help of routing protocols. Each interface of a router belongs to a different network, and a router does not forward broadcasts, so every network connected to it is a separate broadcast domain.
- **Where it is used:** between a LAN and the Internet, between the LANs of an organisation, and throughout the WAN and Internet.

5.10 Gateways

- **What it is:** a device that connects networks which use **different protocols or architectures** and converts between them. A gateway can work at any layer, up to the application layer.
- **Main purpose:** to allow communication between systems that would otherwise be unable to understand each other.
- **How it works:** the gateway receives data in one protocol format, translates it to the protocol format of the other network, and forwards it. It may change address formats, data formats and message structure.
- **Where it is used:** to join dissimilar networks (for example between an e-mail system on one standard and an e-mail system on another), or as the exit point of a network. The term **default gateway** is used for the router a host sends packets to when the destination is outside its own network.

5.11 Comparison of Similar Devices

Hub vs Switch

Feature	Hub	Switch
OSI layer	Physical (layer 1)	Data link (layer 2)
Intelligence	None; does not read addresses	Reads MAC address and keeps a MAC address table
Forwarding	Sends data to all ports	Sends data only to the destination port
Collision domain	One for all ports	One for each port
Bandwidth	Shared among all devices	Dedicated to each port
Duplex mode	Half-duplex	Half- or full-duplex
Security and efficiency	Low; every device sees all traffic	Higher; traffic is directed

Switch vs Router

Feature	Switch	Router
OSI layer	Data link (layer 2)	Network (layer 3)
Address used	MAC (physical) address	IP (logical) address
Connects	Devices within the same network	Different networks
Decision based on	MAC address table	Routing table and best path
Broadcast	Forwards broadcasts to all ports	Does not forward broadcasts (separates broadcast domains)
Typical position	Inside a LAN	At the edge of a LAN, towards other networks or the Internet

Router vs Gateway

Feature	Router	Gateway
Main function	Forwards packets between networks using the same kind of protocol (for example IP)	Converts between different protocols or architectures
OSI layer	Network layer	Any layer up to the application layer
Protocol conversion	No	Yes
Typical example	Home router connecting a LAN to the Internet	E-mail gateway between two mail systems; protocol converter
Note	A router is often used as the default gateway of a LAN	The word gateway is also used loosely for the router at the network exit

Repeater vs Bridge

Feature	Repeater	Bridge
OSI layer	Physical (layer 1)	Data link (layer 2)
Works on	Signals (bits)	Frames and MAC addresses
Function	Regenerates the signal to extend distance	Connects LAN segments and filters traffic
Filtering	None; passes everything	Filters by destination MAC address
Collision domain	Extends one collision domain	Separates collision domains
Error handling	Does not check for errors	Can check frames (for example for errors) before forwarding

Summary of devices and layers

Device	OSI layer	Unit handled	Main role
Connector, transceiver	Physical	Signal	Attach device and cable; send and receive signals
Repeater	Physical	Signal (bits)	Regenerate signal
Hub	Physical	Signal (bits)	Multiport repeater
NIC, PC card	Physical and data link	Frame	Connect host to network
Bridge	Data link	Frame	Join and filter LAN segments
Switch	Data link	Frame	Multiport bridge
Router	Network	Packet	Forward packets between networks
Gateway	Any layer, up to application	Message or packet	Protocol conversion

6. Network Software

The first networks were designed mainly in terms of hardware. Today, network software is highly structured. To reduce design complexity, most networks are organised as a **stack of layers**, each built upon the one below it. The software that makes a network work consists of the protocols and the programs that implement them.

6.1 Layered Architecture

Explanation

- Each **layer** offers certain services to the layer above it and hides the details of how those services are implemented.
- Layer n on one machine carries on a conversation with layer n on another machine. The rules of this conversation are called the **layer n protocol**. The entities that run the layers on different machines are called **peers**.
- The set of operations and services that a lower layer offers to the layer above is described by the **interface** between the layers.
- A list of the protocols used by a system, one per layer, is called a **protocol stack**. A set of layers and protocols is called a **network architecture**.

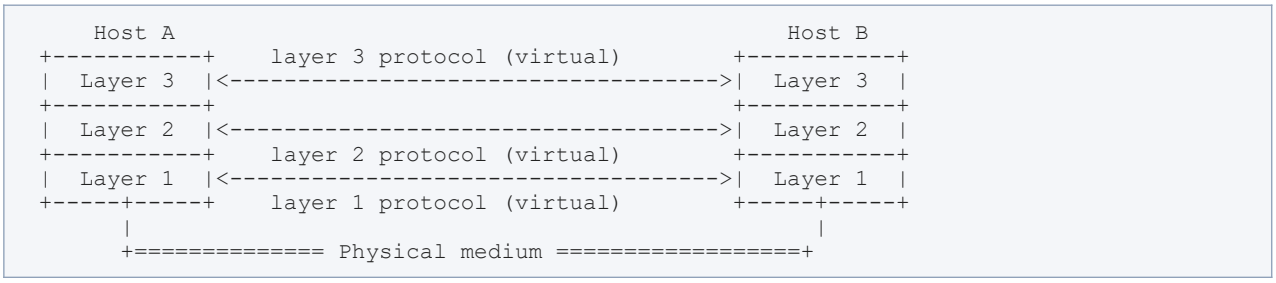


Figure 6.1: Layers, peers and protocols (communication between peers is virtual; real data flows down, across the medium, and up)

In reality no data is transferred directly from layer n of one machine to layer n of the other. Each layer passes the data down to the layer below until the physical medium is reached, and the data then moves up the layers of the receiving machine. On the way down, each layer adds its own control information, called a **header**, to the data it receives from above; at the receiver, each layer removes the header meant for it. This process is called **encapsulation** (and decapsulation at the receiver).

Advantages of layering

- A large problem is divided into smaller, manageable parts.
- A layer can be changed or improved without changing the other layers, provided its interfaces stay the same.
- Equipment and software from different makers can work together when they follow the same layer standards.

6.2 Network Design Issues

The designer of a layered network must decide how the layers will handle a number of common problems. The main design issues are summarised below.

Design issue	Meaning	Typical solution
Addressing	Each message needs the identity of its sender and its receiver.	A unique address for each device or process at the relevant layer.
Error control	Data may be damaged or lost in transmission.	Error-detecting and error-correcting codes; retransmission of lost or damaged data.
Flow control	A fast sender can overload a slow receiver.	Feedback from the receiver telling the sender to slow down or wait.
Congestion control	Too much traffic can overload the network itself.	Limiting the rate at which data is sent when the network is busy.
Sequencing (ordering)	Pieces of a message may arrive in a different order.	Sequence numbers so the receiver can reassemble data in order.
Routing	There may be many possible paths from source to destination.	Choosing a suitable path at each step.
Multiplexing	Several conversations may share one connection.	Combining at the sender and separating at the receiver.
Segmentation and reassembly	A message may be too long for a layer or link to carry in one piece.	Dividing the message into smaller pieces and rebuilding it at the destination.
Reliability	The network should work even when some parts fail.	Alternate paths, error control, redundancy.
Scalability	The design should work as the network grows.	Hierarchical addressing and layered design.
Security	Data should be protected from misuse.	Authentication, confidentiality and integrity protection.
Resource allocation	Network capacity must be shared fairly among users.	Sharing of lines by statistical multiplexing and control of the sending rate.

Some of these issues are dealt with in more than one layer, for example error control and flow control.

6.3 Protocols

Definition

> A **protocol** is a set of rules and conventions that governs the communication between peer entities, that is, between corresponding layers of two communicating systems.

Explanation

A protocol is an agreement between the communicating parties on how communication is to proceed. If two devices use different protocols they cannot understand each other, even if they are physically connected. A protocol has three key elements:

Element	Meaning
Syntax	The structure or format of the data: the order in which the fields are arranged (for example, which part of a frame holds the address and which part holds the data).
Semantics	The meaning of each field, and what action is to be taken based on it.

Element	Meaning
Timing	When data should be sent and how fast, including synchronisation between sender and receiver.

Example

TCP is a protocol of the transport layer, IP of the network layer, and HTTP of the application layer. Ethernet is a protocol for the data link layer.

Protocol vs service

Feature	Service	Protocol
Meaning	The set of operations that a layer offers to the layer above it.	The set of rules used by peer entities to carry out the communication within a layer.
Relates to	Two adjacent layers on one machine (through the interface).	The same layer on two different machines.
Visible to	The user of the layer above.	Only the layer itself; the layer above does not see it.
Freedom	Should remain fixed.	Can be changed without affecting the service, so long as the service is unchanged.

7. Connection-Oriented vs Connectionless Services

A layer can offer two different kinds of service to the layer above it. The choice between them is one of the basic decisions in network design.

7.1 Connection-Oriented Service

Definition

> A **connection-oriented service** is a service in which a connection is first established between the sender and the receiver, data is then transferred over that connection, and the connection is released when the exchange is complete.

Explanation

The service is modelled on the telephone system. To speak to someone, you pick up the phone, dial the number, wait for the other person to answer, talk, and then hang up. In the same way, the service user first sets up a connection, uses it, and then releases it. The connection works like a pipe: data put in at one end comes out at the other end, normally in the same order.

There are three phases:

1. **Connection establishment:** the two sides agree to communicate and may negotiate parameters.
2. **Data transfer:** data flows over the connection, usually with sequence numbers, acknowledgements and flow control.
3. **Connection release:** the connection and the resources reserved for it are freed.

Example

A telephone call; TCP (transport layer); file transfer and e-mail, where every byte must arrive correctly.

7.2 Connectionless Service

Definition

> A **connectionless service** is a service in which each message is sent independently, with its complete destination address, and no connection is set up beforehand.

Explanation

The service is modelled on the postal system. Each letter carries the full address of the receiver and is routed through the system independently of other letters. Two messages sent to the same destination may take different routes and may arrive in a different order. There is no guarantee that the message is delivered unless an additional mechanism is provided.

Example

Ordinary postal mail; UDP (transport layer); IP datagrams; DNS queries and real-time audio or video, where speed is more important than perfect delivery.

7.3 Comparison

Feature	Connection-oriented	Connectionless
Connection establishment	Required before data transfer	Not required
Model	Telephone system	Postal system

Feature	Connection-oriented	Connectionless
Addressing	Full address needed only during set-up; later data uses a connection identifier	Every message carries the full destination address
Path of data	All data follows the same path (route fixed at set-up)	Each message may take a different path
Ordering	Data is delivered in order	Order is not guaranteed
Reliability	Reliable: acknowledgements and retransmission are used	Unreliable (best effort) unless handled by a higher layer
Overhead	Higher: set-up, release, acknowledgements and state to maintain	Lower: no set-up and no state
Speed	Slower to start; steady transfer once connected	Faster for short exchanges
Congestion and resources	Resources can be reserved in advance, so congestion is easier to control	No reservation; harder to control
Failure of a node on the path	The connection is broken	Later messages can use other routes
Examples	TCP, telephone call, virtual circuit	UDP, IP datagram, postal mail
Typical use	File transfer, e-mail, remote login	DNS queries, voice and video streaming, broadcast messages

7.4 Types of Service Quality

Each kind of service can be characterised by its reliability. A reliable service never loses data; it is obtained by having the receiver acknowledge each message. Acknowledgements add overhead and delay, so they are not always worth the cost.

Service	Type	Example
Reliable message stream	Connection-oriented	Sequence of pages, message boundaries preserved
Reliable byte stream	Connection-oriented	Remote login, file download
Unreliable connection	Connection-oriented	Digital voice
Unreliable datagram	Connectionless	Electronic junk mail
Acknowledged datagram	Connectionless	Registered mail, text messaging
Request-reply	Connectionless	Database query

Key Points

- Connection-oriented service is concerned with reliable, ordered delivery; connectionless service is concerned with simple, fast delivery.
- At the network layer, the two ideas are implemented as **virtual circuits** (connection-oriented) and **datagrams** (connectionless).
- In the TCP/IP model, TCP is connection-oriented and UDP is connectionless.

8. Service Primitives

Definition

> A **service** is formally specified by a set of **primitives** (operations) that a user or other entity can invoke to access the service. These operations tell the service to perform some action or to report on an action taken by a peer entity.

Explanation

A service primitive is a basic request that the upper layer makes to the layer below it, or a notice that the lower layer gives to the upper layer. In very simple terms, primitives are the commands, such as "wait for a connection" or "send this data", through which a program uses the services of the network. Each layer provides its service through such a set of primitives.

8.1 Primitives of a Connection-Oriented Service

The following six primitives are used to illustrate a simple connection-oriented service.

Primitive	Meaning	Who uses it
LISTEN	Block (wait) until some process tries to connect.	Server
CONNECT	Actively try to establish a connection with a waiting peer.	Client
ACCEPT	Accept an incoming connection request from a peer.	Server
RECEIVE	Block (wait) for an incoming message.	Either side
SEND	Send a message to the peer.	Either side
DISCONNECT	Terminate the connection.	Either side

8.2 Sequence of Primitives

A client-server interaction proceeds in the following order.

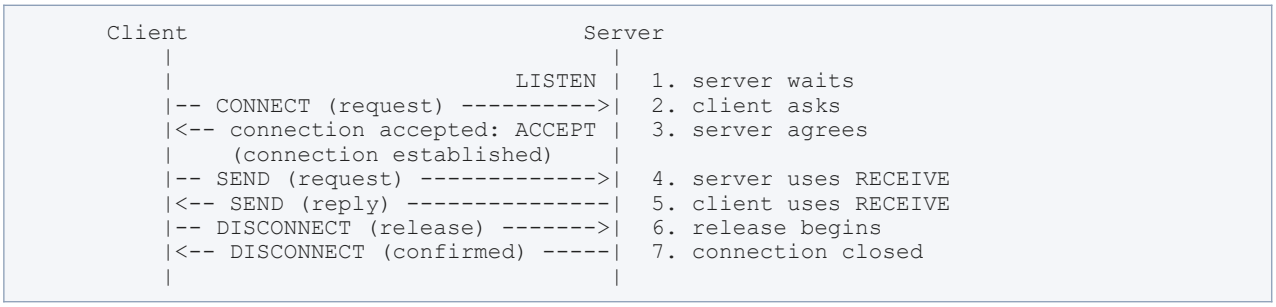


Figure 8.1: Sequence of service primitives in a connection-oriented service

1. The server executes **LISTEN** and waits for a client.
2. The client executes **CONNECT**, which sends a connection request to the server.
3. The server accepts the request (**ACCEPT**) and the connection is established.
4. The client uses **SEND** to transmit a request; the server uses **RECEIVE** to get it.
5. The server replies with **SEND**; the client collects the reply with **RECEIVE**.
6. When the exchange is complete, one side issues **DISCONNECT** and the other side confirms. The connection is released.

In a connectionless service, only the SEND and RECEIVE primitives are needed, because there is nothing to establish or release.

8.3 Classes of Service Primitives

When a service is described formally (as in the OSI model), each primitive is divided into four classes depending on who issues it and in which direction it moves.

Class	Meaning	Direction
Request	A service user asks the service to do something.	User on the sender side to its service provider
Indication	The service informs the user on the other side that an event has occurred.	Service provider to the user on the receiver side
Response	The user on the receiver side answers the indication.	Receiver-side user to its service provider
Confirm	The service informs the original requester of the result.	Service provider to the user on the sender side

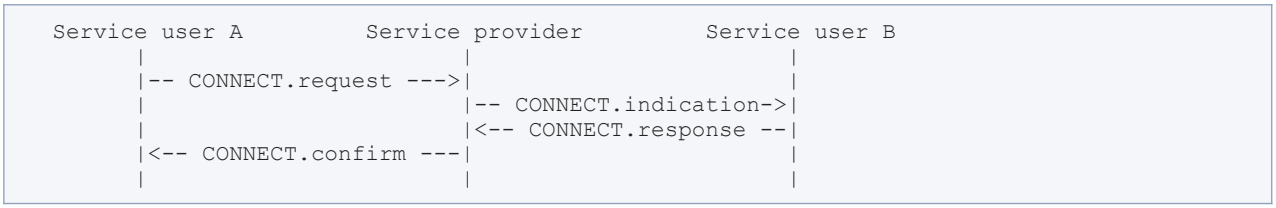


Figure 8.2: The four classes of primitive in a confirmed service (connection establishment)

A service that uses all four classes is a **confirmed service** (connection establishment is always confirmed). A service that uses only request and indication is an **unconfirmed service** (data transfer may be unconfirmed).

Key Points

- Primitives are the interface between a layer and the layer above it; they say what the service does, not how it is done.
- The six basic primitives are LISTEN, CONNECT, ACCEPT, RECEIVE, SEND and DISCONNECT.
- The order is: LISTEN, CONNECT, ACCEPT, data exchange (SEND and RECEIVE), DISCONNECT.

9. OSI Reference Model

Definition

> The **OSI (Open Systems Interconnection) reference model** is a seven-layer model proposed by the International Organization for Standardization (ISO) that describes how data moves from an application on one computer to an application on another computer through a network.

Explanation

The aim of the model is to allow different (open) systems to communicate regardless of their make, design or technology. It divides the work of communication into seven layers, each with a specific function. The OSI model is a **reference model**: it does not define the exact protocols to be used, it only says what each layer should do. The layers are chosen according to these principles: a layer should be created where a different level of abstraction is needed, each layer should perform a well-defined function, the function of each layer should lead to internationally standardised protocols, layer boundaries should be chosen to reduce the flow of information across interfaces, and the number of layers should be large enough to keep functions apart but small enough to remain manageable.

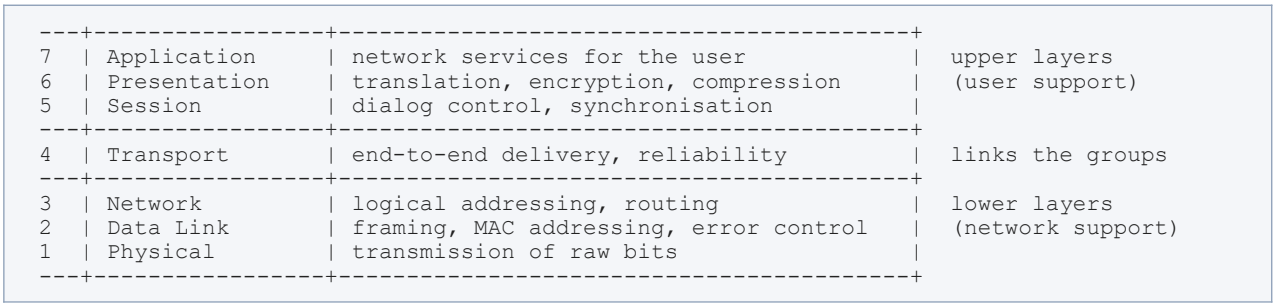


Figure 9.1: The seven layers of the OSI model

9.1 OSI Layer Table

No.	Layer	Data unit	Main function	Examples
7	Application	Data / message	Gives the user or application access to network services	HTTP, FTP, SMTP, Telnet, DNS
6	Presentation	Data / message	Translation, encryption, compression	ASCII, JPEG, MPEG, SSL/TLS (often placed here)
5	Session	Data / message	Establishes, manages and ends sessions; synchronisation	NetBIOS, RPC (examples vary by source)
4	Transport	Segment	End-to-end, process-to-process delivery; reliability, flow control	TCP, UDP
3	Network	Packet	Logical addressing and routing	IP, ICMP; routers
2	Data link	Frame	Framing, physical addressing, error and flow control, access to the medium	Ethernet, PPP, HDLC; bridges, switches, NICs
1	Physical	Bit	Transmission of raw bits over the medium	Cables, connectors, repeaters, hubs

> **Memory aids.** From layer 7 down to layer 1: **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing (Application, Presentation, Session, Transport, Network, Data link, Physical). From layer 1 up to layer 7: **P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way.

9.2 Layer 1: Physical Layer

- **Main purpose:** to transmit raw bits (0s and 1s) as signals over the transmission medium.
- **Important functions:**
  - Defines the electrical, mechanical and functional characteristics of the interface and the medium (cables, connectors, voltages).
  - Representation of bits: how 0s and 1s are encoded as signals.
  - Data rate: the number of bits sent per second.
  - Synchronisation of the sender and receiver clocks at the bit level.
  - Defines the physical topology and the transmission mode (simplex, half-duplex, full-duplex).
- **Examples (protocols and devices):** cables and connectors, repeaters, hubs, transceivers; standards such as RS-232.
- **Unit of data:** bit
- **Easy way to remember:** the "wires and signals" layer: it carries bits and knows nothing about their meaning.

9.3 Layer 2: Data Link Layer

- **Main purpose:** to deliver data reliably from one node to the next node (hop-to-hop or node-to-node) over a physical link, and to turn the raw bit stream into error-free frames.
- **Important functions:**
  - Framing: dividing the bit stream into manageable units called frames.
  - Physical (MAC) addressing: adding the address of the sender and receiver to each frame.
  - Error control: detecting damaged or lost frames and retransmitting them.
  - Flow control: keeping a fast sender from swamping a slow receiver.
  - Access control: deciding which device may use the shared link at a given time (MAC sublayer). The upper part, the LLC sublayer, deals with logical link control.
- **Examples (protocols and devices):** Ethernet, PPP, HDLC; NICs, bridges and switches work at this layer.
- **Unit of data:** frame
- **Easy way to remember:** "frames between neighbours": it moves a frame to the very next device on the link.

9.4 Layer 3: Network Layer

- **Main purpose:** to deliver packets from the source host to the destination host, possibly across many networks (source-to-destination delivery).
- **Important functions:**
  - Logical addressing: adding the logical (IP) addresses of sender and receiver.
  - Routing: finding the path from source to destination across interconnected networks.
  - Packet forwarding between networks.
  - Congestion control in the subnet; fragmentation of packets where needed.
- **Examples (protocols and devices):** IP, ICMP, routing protocols; routers work at this layer.
- **Unit of data:** packet
- **Easy way to remember:** "routing and roads": it chooses the road for the packet from end to end.

9.5 Layer 4: Transport Layer

- **Main purpose:** to deliver the complete message from one process on the source host to a process on the destination host (end-to-end or process-to-process delivery), with the level of reliability required.
- **Important functions:**
  - Service-point (port) addressing: identifying the correct process on the host.
  - Segmentation and reassembly: dividing the message into segments with sequence numbers and rebuilding it.
  - Connection control: connection-oriented (TCP) or connectionless (UDP) delivery.
  - Flow control and error control between the two end points.
  - Multiplexing of several conversations onto one network connection.
- **Examples (protocols and devices):** TCP, UDP.
- **Unit of data:** segment
- **Easy way to remember:** "the delivery guarantee": the transport layer makes sure the whole message reaches the right program.

9.6 Layer 5: Session Layer

- **Main purpose:** to establish, maintain and terminate sessions (dialogues) between communicating applications.
- **Important functions:**
  - Dialog control: decides whether communication is half-duplex or full-duplex and whose turn it is to send.
  - Synchronisation: inserts checkpoints into the data stream so that, after a failure, transfer can resume from the last checkpoint instead of from the beginning.
  - Token management: prevents two sides from attempting the same critical operation at the same time.
- **Examples (protocols and devices):** NetBIOS and RPC are sometimes quoted as examples; the TCP/IP model has no separate session layer. Specific protocols vary with the reference text.
- **Unit of data:** data (message)
- **Easy way to remember:** "conversation manager": it opens, controls and closes the conversation.

9.7 Layer 6: Presentation Layer

- **Main purpose:** to deal with the syntax and semantics of the information exchanged, so that data sent by one system can be understood by another.
- **Important functions:**
  - Translation: converting between different character codes and data formats.
  - Encryption and decryption for privacy.
  - Compression to reduce the number of bits transmitted.
- **Examples (protocols and devices):** ASCII and Unicode text, JPEG, GIF and MPEG formats; SSL/TLS is often associated with this layer.
- **Unit of data:** data (message)
- **Easy way to remember:** "the translator and packager": it changes the form of the data, not its meaning.

9.8 Layer 7: Application Layer

- **Main purpose:** to give the user (a person or a program) access to the network; it is the layer closest to the user.
- **Important functions:**
  - Network virtual terminal: remote log-in to another computer.
  - File transfer, access and management.
  - Mail services: e-mail forwarding and storage.
  - Directory services and access to the World Wide Web.
- **Examples (protocols and devices):** HTTP, FTP, SMTP, Telnet, DNS.
- **Unit of data:** data (message)
- **Easy way to remember:** "what the user sees": it provides the services that applications such as a browser or mail program use.

9.9 Data Flow and Encapsulation in the OSI Model

At the sending host, data starts at the application layer and moves down. Each layer adds its own header (the data link layer also adds a trailer) and treats everything it received from above as data. At the receiving host the process is reversed: each layer removes its header and passes the rest upward.

```
Layer            Unit passed down
7 Application    [H7][ DATA ]
6 Presentation   [H6][H7][ DATA ]
5 Session        [H5][H6][H7][ DATA ]
4 Transport      [H4][H5][H6][H7][ DATA ]            -> segment
3 Network        [H3][H4][H5][H6][H7][ DATA ]        -> packet
2 Data Link      [H2][H3][...][ DATA ][T2]           -> frame
1 Physical       0 1 1 0 1 0 0 1 ...                 -> bits on the medium

Hn = header added by layer n      T2 = data link trailer
```

Figure 9.2: Encapsulation of data as it moves down the layers

Types of delivery

Layer	Delivery responsibility
Data link	Node-to-node (hop-to-hop) delivery of frames
Network	Source-to-destination (host-to-host) delivery of packets
Transport	Process-to-process (end-to-end) delivery of the complete message

Key Points

- OSI is a reference model, not a set of protocols.
- Layers 1 to 3 are the network support layers; layers 5 to 7 are the user support layers; layer 4 links the two groups.
- The data unit is bit, frame, packet and segment at layers 1 to 4.

10. TCP/IP Model

Definition

> The **TCP/IP model** (also called the TCP/IP reference model or Internet model) is the layered protocol architecture of the Internet. It is named after its two main protocols, the **Transmission Control Protocol (TCP)** and the **Internet Protocol (IP)**.

Explanation

The TCP/IP model grew out of the ARPANET, a research network sponsored by the U.S. Department of Defense. A main design goal was that connections between machines should remain intact even if some part of the network was destroyed. In contrast with OSI, the protocols came first and the model was a description of the protocols that already existed. The model has four layers. Some textbooks split the lowest layer into physical and data link layers and describe five layers; the four-layer form is used in these notes.

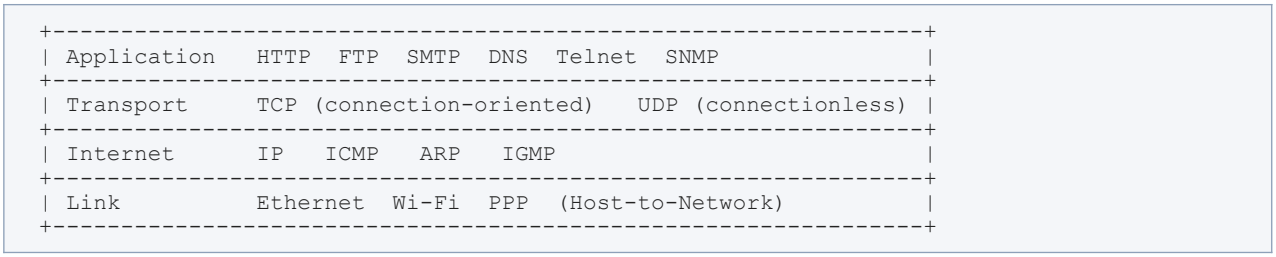


Figure 10.1: Layers and typical protocols of the TCP/IP model

10.1 Layers of the TCP/IP Model

Layer 1: Host-to-Network (Link) Layer

- **Purpose:** to connect the host to the network and send IP packets over the local link. The model does not define a specific protocol for this layer; the host uses whatever protocol suits the physical network.
- **Functions:** framing, physical addressing and the actual transmission of bits over the medium. It corresponds to the physical and data link layers of the OSI model.
- **Examples:** Ethernet, Wi-Fi, PPP.
- **Data unit:** frame.

Layer 2: Internet Layer

- **Purpose:** to allow hosts to inject packets into any network and have them travel independently to the destination, possibly through different networks. Packets may arrive in a different order than they were sent, so higher layers must reorder them if required.
- **Functions:** logical (IP) addressing, routing and forwarding of packets, fragmentation. It is based on a connectionless packet-switching service.
- **Protocols: IP** (Internet Protocol) defines the packet format and delivers packets on a best-effort basis; **ICMP** reports errors and gives control messages; **ARP** maps an IP address to a physical (MAC) address; **IGMP** supports group communication. ARP is commonly grouped with this layer.
- **Data unit:** packet (datagram). Corresponds to the network layer of OSI.

Layer 3: Transport Layer

- **Purpose:** to allow peer entities on the source and destination hosts to carry on a conversation (process-to-process delivery).

- **Functions:** port addressing, segmentation, multiplexing, and, for TCP, reliability, flow control and congestion control.
- **Protocols: TCP** is a reliable, connection-oriented protocol that delivers a byte stream without error; **UDP** is an unreliable, connectionless protocol used where prompt delivery is more important than accuracy, such as voice, video and simple queries.
- **Data unit:** segment (TCP) or user datagram (UDP). Corresponds to the transport layer of OSI.

Layer 4: Application Layer

- **Purpose:** to provide network services directly to the user and to applications. It contains all the higher-level protocols; there are no separate session or presentation layers.
- **Functions:** user interface to the network, data formatting and encoding, and dialogue handling are all performed by the application protocols.
- **Protocols:** HTTP (web), FTP (file transfer), SMTP (e-mail), DNS (names to IP addresses), Telnet (remote login), SNMP (network management).
- **Data unit:** message. Corresponds to the session, presentation and application layers of OSI.

10.2 Encapsulation in TCP/IP

As in OSI, each layer adds its own header: the application data is placed in a TCP or UDP segment, the segment is placed in an IP packet, and the packet is placed in a frame of the link layer, which is sent as bits.

```
Application   [ DATA ]
Transport     [TCP hdr][ DATA ]                  segment
Internet      [IP hdr][TCP hdr][ DATA ]          packet
Link          [Frame hdr][IP][TCP][ DATA ][Trl]  frame
```

Figure 10.2: Encapsulation in the TCP/IP model

11. OSI vs TCP/IP

11.1 Relationship Between the Two Models

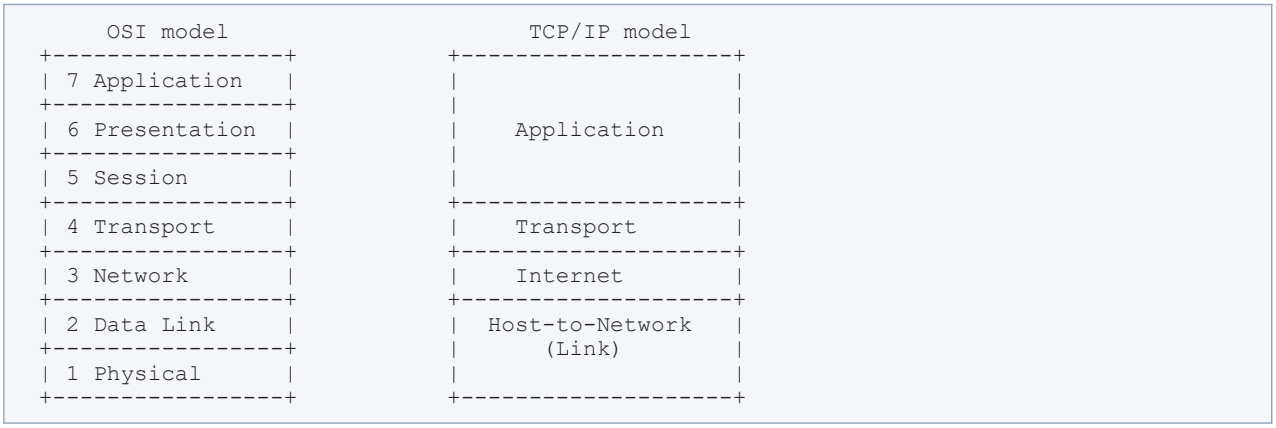


Figure 11.1: Mapping of OSI layers to TCP/IP layers

11.2 Comparison Table

Feature	OSI model	TCP/IP model
Full form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Developed by	ISO (International Organization for Standardization)	U.S. Department of Defense (ARPANET research)
Number of layers	Seven	Four (some texts use five)
Approach	Model first, protocols afterwards (protocol independent)	Protocols first, model describes them
Session and presentation layers	Separate layers	Not present; their work is done by the application layer
Physical and data link layers	Two separate layers	Combined in the host-to-network (link) layer
Services, interfaces, protocols	Clearly distinguished	Not clearly distinguished
Network layer service	Both connectionless and connection-oriented	Connectionless only (Internet layer)
Transport layer service	Connection-oriented only	Both connection-oriented (TCP) and connectionless (UDP)
Generality	Very general; can describe other protocol stacks	Not general; describes only the TCP/IP protocols
Reliability of transport	Guaranteed by the transport layer	TCP is reliable, but UDP is not
Use in practice	Used as a reference and teaching model; its protocols are little used	Used on the Internet; the protocols are widely deployed

11.3 Similarities

- Both are based on the idea of a stack of independent protocols arranged in layers.
- The functions of the layers are broadly similar: the layers up to and including the transport layer provide an end-to-end, network-independent transport service for processes that want to communicate.
- In both, the layers above the transport layer are application-oriented users of the transport service.

Important Definitions

Term	Definition
Data communication	The exchange of data between two or more devices through a transmission medium.
Computer network	A collection of autonomous computers interconnected by a single technology so that they can exchange information and share resources.
Protocol	A set of rules and conventions that governs communication between peer entities; its elements are syntax, semantics and timing.
LAN	A privately owned network connecting devices within a limited area such as a room, building or campus.
MAN	A network covering a city or large town, often joining several LANs.
WAN	A network covering a large geographical area such as a country or continent, made of hosts and a communication subnet.
Network topology	The geometric arrangement of the nodes and links of a network.
Simplex, half-duplex, full-duplex	One-way only; both ways but one at a time; both ways at the same time.
Connector	A component that joins a cable to a device or to another cable.
Transceiver	A device combining a transmitter and a receiver in one unit.
Repeater	A physical layer device that regenerates a signal to extend the distance it can travel.
Hub	A multiport repeater that sends incoming data to all other ports.
NIC	The hardware that connects a computer to a network and carries its unique MAC address.
PC card	A credit-card-sized expansion card (PCMCIA) that adds features such as network access to a laptop.
Bridge	A data link layer device that connects LAN segments and filters frames by MAC address.
Switch	A multiport bridge that forwards frames only to the port of the destination device.
Router	A network layer device that forwards packets between networks using IP addresses and a routing table.
Gateway	A device that connects networks with different protocols and converts between them.
Layer, interface, peer	A layer offers services to the one above; the interface separates adjacent layers; peers are the same-layer entities on two machines.
Connection-oriented service	A service in which a connection is set up, used and released; data is delivered reliably and in order.
Connectionless service	A service in which each message is sent independently with its full address and no connection is set up.
Service primitive	An operation that a user can invoke to access a service of a layer (for example LISTEN, CONNECT, SEND).
OSI reference model	A seven-layer model proposed by ISO describing how data moves between applications on different systems.
TCP/IP model	The four-layer architecture of the Internet named after the TCP and IP protocols.

Term	Definition
Encapsulation	The addition of a header (and sometimes a trailer) by each layer to the data received from the layer above.

Important Comparisons

The following comparisons are frequently asked. Each is covered by a table in these notes.

Comparison	Where to find it
LAN vs MAN vs WAN	Section 3.4
Bus vs Star vs Ring vs Mesh vs Tree topology	Section 4.7
Hub vs Switch	Section 5.11
Switch vs Router	Section 5.11
Router vs Gateway	Section 5.11
Repeater vs Bridge	Section 5.11
Service vs Protocol	Section 6.3
Connection-oriented vs Connectionless service	Section 7.3
Simplex vs Half-duplex vs Full-duplex	Section 1.1
Data communication vs Computer network	Section 1.3
OSI model vs TCP/IP model	Section 11.2

Important Exam Points

- A data communication system has five components: message, sender, receiver, medium and protocol.
- A computer network is a collection of **autonomous** computers interconnected by a single technology.
- Networks are classified by scale: LAN (building or campus), MAN (city), WAN (country or continent).
- A WAN consists of hosts and a communication subnet made of transmission lines and routers.
- Bus uses one shared backbone with terminators; star uses a central hub or switch; ring is a closed loop; full mesh needs n(n-1)/2 links.
- The central device is the single point of failure in a star topology, and the backbone cable in a bus topology.
- Repeater and hub work at the physical layer; bridge and switch at the data link layer; router at the network layer; gateway can work at any layer.
- A hub broadcasts to all ports (one collision domain); a switch forwards to the destination port only (a collision domain per port).
- A switch uses the MAC address; a router uses the IP address.
- A repeater regenerates a signal; an amplifier amplifies the signal together with the noise.
- A NIC holds a unique 48-bit MAC address.
- Layering hides implementation details; peers communicate through a protocol; adjacent layers communicate through an interface.
- A protocol has three elements: syntax, semantics and timing.
- Main design issues: addressing, error control, flow control, congestion control, sequencing, routing, multiplexing, security.
- Connection-oriented service has three phases (establishment, transfer, release) and is modelled on the telephone; connectionless service is modelled on the postal system.
- The six basic primitives are LISTEN, CONNECT, ACCEPT, RECEIVE, SEND, DISCONNECT. The four classes are request, indication, response and confirm.
- OSI layers from the top: Application, Presentation, Session, Transport, Network, Data link, Physical.
- Data units: bit (physical), frame (data link), packet (network), segment (transport).
- Delivery: data link is node-to-node, network is host-to-host, transport is process-to-process.
- TCP/IP layers from the top: Application, Transport, Internet, Host-to-Network (Link).
- OSI supports both service types at the network layer but only connection-oriented at the transport layer; TCP/IP has connectionless service at the Internet layer and both at the transport layer.

Quick Revision

Topic	Summary
Data communication and networks	Data communication is the transfer of data between devices over a medium under a protocol. A computer network is a set of autonomous interconnected computers that share information and resources.
Uses	Business: resource sharing, client-server, communication, e-commerce. Home: information, communication, entertainment, shopping. Mobile users use wireless access. Social issues include privacy and security.
Types	LAN is small, private and fast. MAN covers a city. WAN covers countries and is built from hosts plus a subnet of routers.
Topologies	Bus (shared cable), star (central device), ring (loop), mesh (all pairs linked), tree (hierarchy of stars), hybrid (combination).
Hardware	Connectors and transceivers attach devices to cables. Repeater and hub: layer 1. NIC and PC card connect hosts. Bridge and switch: layer 2, use MAC addresses. Router: layer 3, uses IP addresses. Gateway: converts protocols.
Network software	Networks are layered. Each layer gives services to the layer above and talks to its peer by a protocol. Design issues include addressing, error, flow and congestion control, routing, multiplexing and security.
Services	Connection-oriented: set-up, transfer, release; reliable and ordered; like the telephone. Connectionless: independent messages; fast; like the post.
Primitives	LISTEN, CONNECT, ACCEPT, RECEIVE, SEND, DISCONNECT. Request, indication, response and confirm describe how a service is invoked and answered.
OSI model	Seven layers: physical (bits), data link (frames), network (packets, routing), transport (end-to-end), session (dialog), presentation (syntax, encryption, compression), application (user services).
TCP/IP model	Four layers: link, Internet (IP), transport (TCP and UDP), application. No session and presentation layers.
OSI vs TCP/IP	OSI has seven layers and is a general reference model; TCP/IP has four layers and describes the protocols of the Internet.

Possible Exam Questions

A. Short-Answer Questions

1. Define data communication. List its five components.
2. What is a computer network? State any two uses of computer networks.
3. Define LAN, MAN and WAN.
4. What is meant by network topology? Differentiate between physical and logical topology.
5. What is the function of a repeater?
6. What is a NIC? What is a MAC address?
7. What is a PC card? Where is it used?
8. What is a protocol? Name its three elements.
9. What are service primitives? List the six basic primitives.
10. Name the seven layers of the OSI model in order.
11. Name the layers of the TCP/IP model.
12. What is encapsulation?

B. Conceptual Questions

1. Explain the client-server model with a neat diagram.
2. Explain the uses of computer networks for business and for home users.
3. Why is a layered architecture used in computer networks? Explain peers, interfaces and protocol stack.
4. Explain the main design issues for the layers of a network.
5. Explain how a connection-oriented service works. Give its three phases.
6. Explain the working of the primitives LISTEN, CONNECT, ACCEPT, SEND, RECEIVE and DISCONNECT in a client-server interaction.
7. Explain request, indication, response and confirm primitives with a diagram.
8. Explain how a bridge learns addresses and decides whether to filter or forward a frame.
9. Explain the function of the transport layer and how it differs from the network and data link layers.
10. Explain the layers of the TCP/IP model with their protocols.

C. Difference and Comparison Questions

1. Differentiate between LAN, MAN and WAN.
2. Compare bus, star, ring and mesh topologies.
3. Differentiate between a hub and a switch.
4. Differentiate between a switch and a router.
5. Differentiate between a router and a gateway.
6. Differentiate between a repeater and a bridge.
7. Differentiate between connection-oriented and connectionless services.
8. Differentiate between a service and a protocol.
9. Compare the OSI reference model and the TCP/IP model.
10. Differentiate between data communication and a computer network.

D. Long-Answer Questions

1. Describe the types of computer networks (LAN, MAN, WAN) with their characteristics, examples and a comparison table.
2. Explain the different network topologies with neat diagrams, giving the advantages and disadvantages of each.
3. Describe the network hardware components: connectors, transceivers, repeaters, hubs, NICs, PC cards, bridges, switches, routers and gateways.
4. Explain the OSI reference model. Describe the functions of each of its seven layers and show how data moves through the layers.
5. Explain the TCP/IP model with its layers and protocols. Compare it with the OSI model and show the mapping between their layers.
6. Explain network software: layered architecture, protocols and the network design issues. Explain how connection-oriented and connectionless services differ and how service primitives are used.